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Palaeomagnetic evidence for block rotations in the western Greek Rhodope

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SUMMARY

The Rhodope massif has been traditionally considered to be a pre-Mesozoic element that remained stable during the Tertiary. This idea has been partly supported by previous palaeomagnetic results on plutonic and volcanic rocks ranging in age from Palaeocene to Oligo-Miocene. However, new palaeomagnetic data from additional sites on Tertiary magmatic rocks of the western Greek Rhodope show declination values indicating, in most cases, a clockwise rotation, whereas inclinations are systematically lower than those corresponding to the actual latitudinal position of the Rhodope. Our results, together with previously published ones, are discussed in the light of recent ideas proposed for the Rhodope and the north Aegean region. We suggest that at least the western part of the Greek Rhodope has been clockwise rotated after the Oligocene. North–south to NE–SW extension tilted the formations towards the north or north-east, producing apparently high inclinations, if the alternative pole position for Eurasia proposed by Westphal (1990) is adopted.

Key words: block rotation, Greek Rhodope, magmatic rocks, palaeomagnetism.

INTRODUCTION

There are still controversies concerning the origin, the age and the geodynamic role of the Rhodope massif of northern Greece. The old ideas, however, that the Rhodope has been immobile since its consolidation in the Precambrian or Late Palaeozoic (Kober 1931; Jaranov 1938; Aubouin 1960) do not hold today. More recent findings support the view that the Rhodope was actually a fold and thrust belt magmatically active during the Late Cretaceous and Early Tertiary, which has been affected by extensional collapse and tectonic unroofing in the Middle Miocene to Early Pliocene (Ivanov 1981, 1988; Papanikolaou & Panagopoulos 1981; Zachos & Dimadis 1983; Maltezou & Brooks 1989; Burgh *et al.* 1990; Kilias & Mountrakis 1990; Koukouvelas & Doutsos 1990; Kolokotroni & Dixon 1991; Kolokotroni 1992; Jones *et al.* 1992; Dinter & Royden 1993).

Late Cretaceous and Early Tertiary thrusting in the Rhodope may have been the result of shortening due to continental collisions eliminating Tethyan (or Palaeotethyan) strands left between the northward drifted Gondwanian fragments, or between them and the Eurasian margin. The provenance of Rhodope (Eurasian or Gondwanian) is still controversial.

With only the exception of a small granitic body of Late Oligocene age (granite of Leptokarya, Spais 1987) which showed a 23° eastward declination, previous palaeomagnetic studies in the Rhodope (Table 1) did not reveal Tertiary rotations (Kissel *et al.* 1986; Spais 1987; Kondopoulou & Pavlides 1990; Nozharov, Dolapchieva & Petkov 1990). More recent palaeomagnetic measurements, however, questioned the view of an unrotated Rhodope and showed that the clockwise rotation seen in the Leptokarya granite is not an exception. For this reason a more detailed palaeomagnetic study of Tertiary igneous rocks intruded or extruded in the western Greek Rhodope was carried out, in order to determine the magnitude, and if possible the mechanism, of rotation(s) affecting the Rhodope since the Early Tertiary. The first results of this study are summarized in this paper.

SAMPLING

Sampling for the present study covers the Greek part of the Rhodope between 23°E and 25°E and from 40.8°N to 41.4°N. This area, as well as more northerly parts of the Rhodope in Bulgaria, is characterized by a widespread Oligocene to Early Miocene mainly plutonic, igneous activity (see also Dürr *et al.* 1978; Maltezou & Brooks 1989; Jones *et al.* 1992). Cores from the Tertiary plutonic and

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Table 1. Published palaeomagnetic results for the broader area.

AREA	TYPE OF ROCKS	AGE	N (SITES)	D	I	κ	a95	Lat.	Long.	REFERENCES
FERES-ESSIMI-LEPTOKARYA-MARONIA	VOLC.-PLUT.	32.5 Ma.	15	7	46.5	23	7.5	-	-	1. Kissel et al. (1986)
ESSIMI	VOLCANICS	23-35 Ma.	-	2.3	52.1	-	6	81.5	192.3	2a. Spais (1987)
LEPTOKARYA	PLUTONITES	23-35 Ma.	12	22.8	46.4	-	4.5	67.05	324.3	2b. Spais (1987)
SERRES	PLUTONITES	29-33 Ma.	2	8	50	26	19	-	-	3. Kondopoulou (1986)
KILKIS	VOLCANICS	23-28 Ma.	5	329	48	46	19	-	-	4. Pavlides et al. (1988)
CHALKIDIKI	PLUTONITES	35±5 Ma.	10	37	31	28	9	50.0	139.0	5. Kondopoulou & Westphal (1986)
LEMNOS	VOLCANICS	17-21 Ma.	7	34	48	18	15	-	-	6. Westphal & Kondopoulou (1989)
BULGARIAN RHODOPE	VOLCANICS	30-39 Ma.	-	355.4	57.3	-	7.6	84.9	250.6	7. Dolapchieva et al. (1986)
E.RHODOPE	VOLCANICS	OLIGOCENE	36	357.4	59.2	13.1	6.8	87	258	8. Nozharov et al. (1990)
STRIMONIKO	VOLCANICS	28 Ma.	3	358	58	355	7	-	-	9. Westphal et al. (1991)
METALLIKO	VOLCANICS	EOCENE	5	39	33	16	19	-	-	10. Westphal et al. (1991)

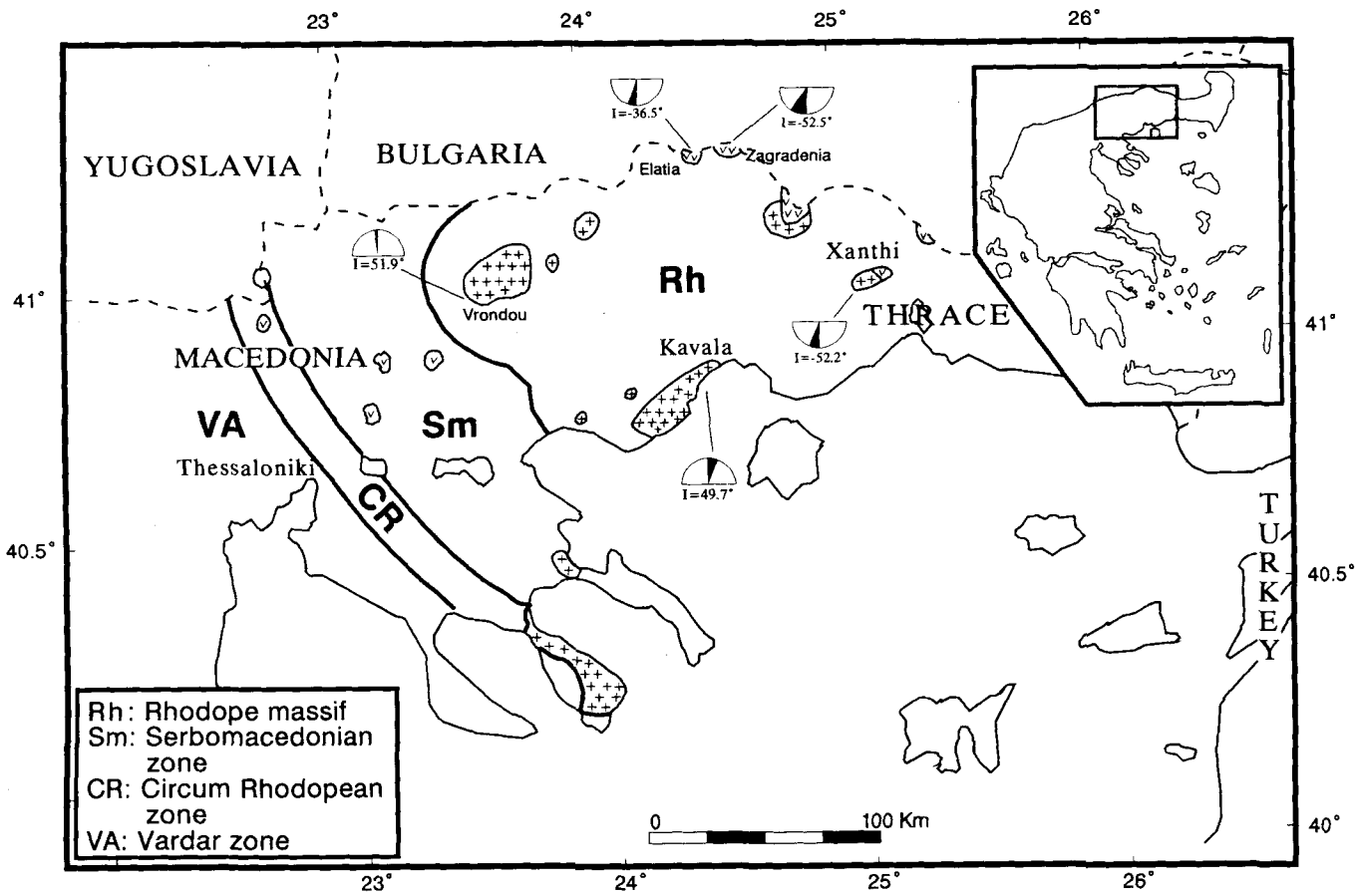


Figure 1. Sampling locations and palaeomagnetic results of the studied area. (Numbers represent the *I* values obtained, thick lines indicate the limits of the geotectonic zones.)

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Table 2. Palaeomagnetic results obtained in the present study.

AREA	Lat. λ_s	Long. φ_s	TYPE OF ROCKS	AGE	N (SITES)	D	I	κ	α_{95}	δD	δI	Lat. λ_p	Long. φ_p
XANTHI	41.15	24.88	PLUTONITES	28 Ma	9	208.4	-53.5	29.3	9.5	15.97	9.5	66.5	123.3
ELATIA	41.5	24.34	VOLCANICS	OLIGOCENE	5	194.5	-38.5	42.4	11.9	15.20	11.9	66.7	168.3
ELATIA1	41.5	24.34	PLUTONITES	86±27Ma	1	210.1	-41.5	-	-	-	-	-	-
ELATIA2 (ZAGRADENIA)	41.5	24.4	VOLCANICS	OLIGOCENE	2	212.8	-51.2	-	-	-	-	-	-
VRODOU	41.3	23.57	PLUTONITES	29-33 Ma	1	191.9	-55.3	-	-	-	-	-	-
KAVALA	40.9	24.0	PLUTONITES	18 Ma	8	24.3	53.2	22.5	10.9	18.2	10.9	69.5	126.5

Mean Direction: $D_m=23.7$, $I_m=49.1$, $N=6$, $k=80.7$, $\alpha_{95}=7.5°$

volcanic rocks cropping out in the studied area were sampled and measured (Fig. 1). The sampling sites were selected on the basis of material freshness, the morphology and the lack of intense post-emplacement tectonism.

Sampling was carried out with a portable drill and cores were oriented with both magnetic and sun compasses. A total of 250 cores from 28 sites were collected for this study. Table 2 provides information about the sampled igneous bodies, their age, the location of the sampling sites and the rock types.

MEASUREMENTS AND RESULTS

Demagnetization processes and component analysis

The NRM has been measured in most cases with a MOLSPIN spinner magnetometer and in some cases with a modified Digico magnetometer. The lowest NRM intensities measured were about 0.2 A m^{-1}, the highest ones reaching in some cases 1.5 A m^{-1}.

The magnetic cleaning was carried out with an AF demagnetizer (University of Thessaloniki) and with thermal treatment (University of Strasbourg). In the first case the field usually reached 70 mT (in a few cases 100 mT) before the stable component was destroyed. In the second case most of the samples were completely demagnetized at about 580 °C; for a limited number of them, however, the temperature had to reach 650 °C. During the thermal treatment, the evolution of magnetic susceptibility was checked on pilot samples; in most cases no significant variation was registered. The results show important changes at the sites EL7, 9, 10. The variation of K for the other sites is practically insignificant.

Zijderveld diagrams were drawn using standard palaeomagnetic techniques. In the majority of cases the primary component was calculated with high accuracy while some samples have been rejected because of their anomalous behaviour during the demagnetization.

Demagnetization revealed three types of behaviour (Fig. 2).

(1) The presence of a single resistant component, in the majority of cases (e.g. EL9-01B; Fig. 2a).

(2) The presence of a small secondary component (viscous), easily destroyed at 15 mT or 180 °C. (e.g. KA1-3A; Fig. 2b).

(3) The presence of a more resistant secondary component, destroyed at about 300 °C. (e.g. XA4-09A, $D = 322°$ and $I = 6.5°$; Fig. 2c).

The stereographic projections show that the within-site scatter is very small. Also noticeable is the existence of both normal and reverse polarities, which are almost antipodal, though the reverse ones are more numerous (Fig. 3). This is an indication of a satisfactory cleaning and stability of magnetization.

The results obtained appear in Table 2.

Magnetic mineralogy

Thin sections of the measured samples (one sample per site) have been examined under a reflecting light microscope. Magnetite and, mainly secondary, haematite are the main opaque constituents in the rocks studied. Their relative proportions, which do not vary much between the different sites of the same area, were visually estimated and are given below.

Xanthi: magnetite 90 per cent haematite 10 per cent;
Kavala: magnetite 95 per cent, haematite 5 per cent;
Elatia: magnetite 90–95 per cent, haematite 5–10 per cent;
Zagradenia: haematite 100 per cent;
Vrondou: magnetite 50 per cent, haematite 50 per cent.

In most cases the grains are distinct. The grain size of the magnetic carrier has been measured for each examined thin section. Grain diameters do not exceed 0.06 mm and the majority of grains are in the range 0.02–0.04 mm. Grains can thus be considered as sufficiently small to reflect accurately the geomagnetic field direction (Tarling 1983).

Stepwise acquisition of the isothermal remanent magnetization (IRM) indicates that in most cases saturation is

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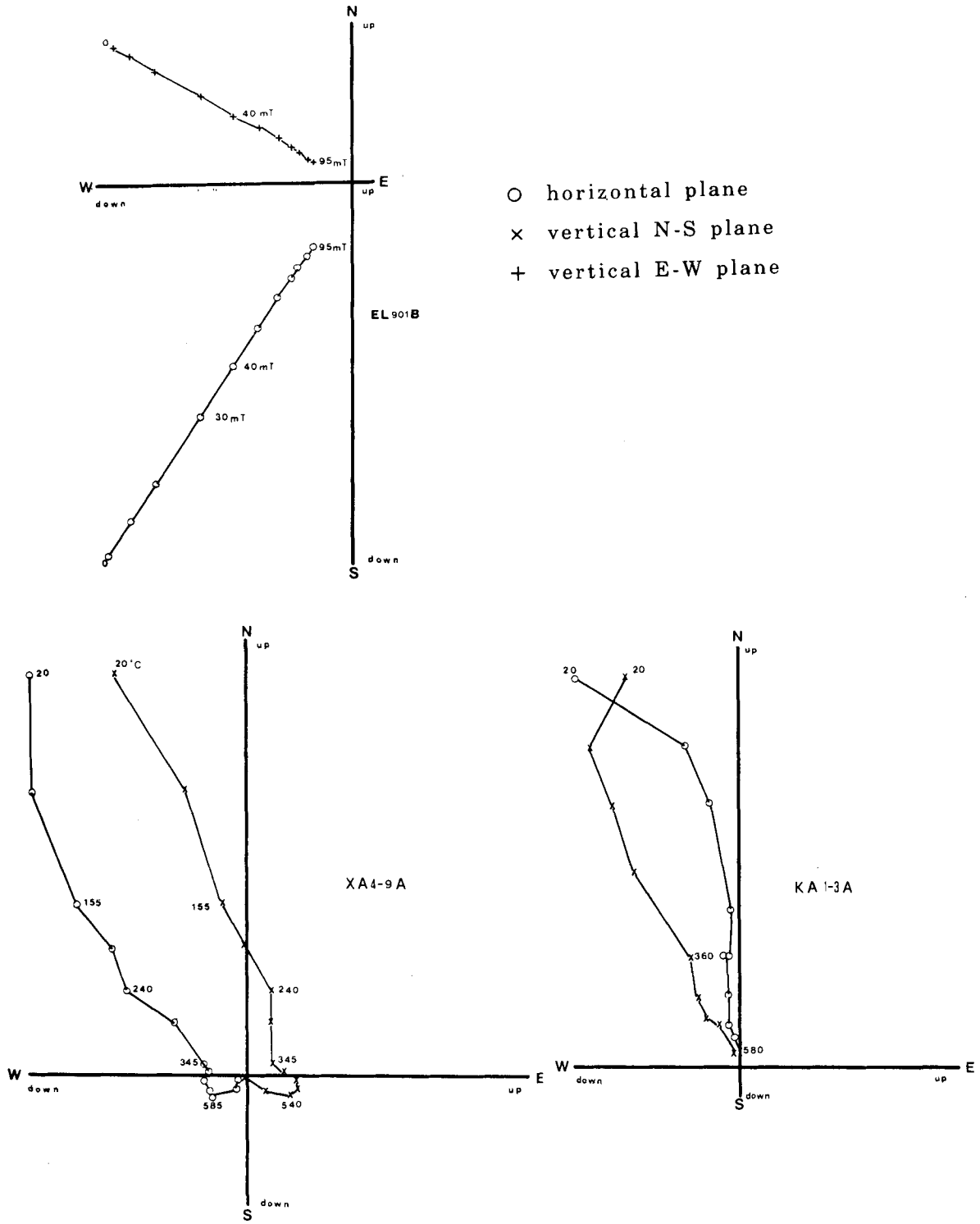


Figure 2. Examples of AF (EL9 01B) and thermal demagnetizations (cases KA1 3A and XA4 09A).

reached in fields lower than 0.02 T (Fig. 4). The dominance of magnetite as the main magnetic carrier can thus be assumed (with the exception of Zagradenia).

A thermomagnetic analysis on pilot samples has been performed by study of the decrease in saturation magnetization as a function of temperature in field-free space. The I_{rs} versus T curves recorded during the first heating of each specimen to 650 °C give the blocking temperatures T_b of the magnetic minerals in a rock in its natural state. The curves recorded during the second heating show the influence of the thermal treatment on the rock, resulting in changes in I_{rs} and in T_b spectrum (Kadziatko-Hofmokl & Kruczyk 1976). Typical examples of the curves obtained are shown in Fig. 5, and indicate the presence of

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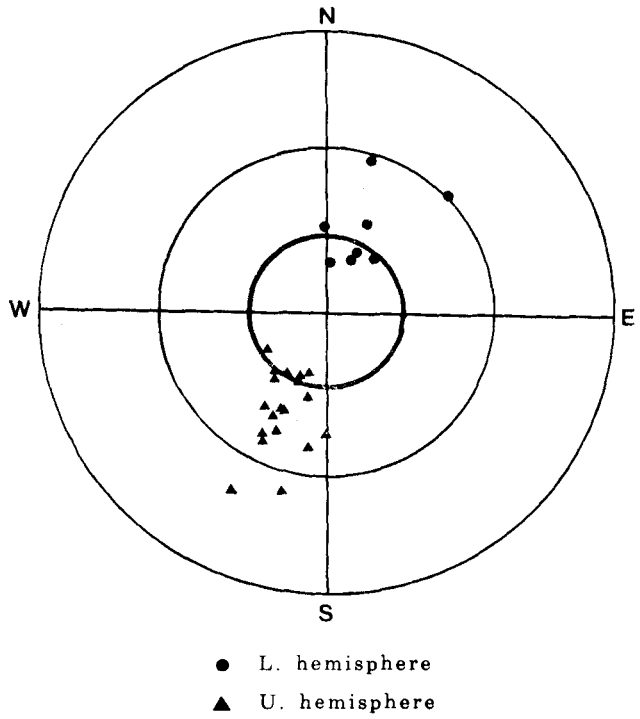


Figure 3. Stereographic projection of mean directions for all sites studied.

magnetite and small amounts of haematite in most of the cases.

In summary, the magnetic mineralogy experiments show that the sampled formations are suitable for a palaeomagnetic study.

EVALUATION OF THE RESULTS

A major problem in our study was the lack of structural corrections due to the nature of the intrusive units sampled, as the palaeohorizontal is difficult to assess for granites. We partially bypassed this problem by considering the consistency between the site mean directions over a broad area (20 000 km^2) and also the similarity of the results obtained from the volcanics in Zagradenia. An additional support of this consistency was the similarity of these mean directions with the ones obtained in the Serbomacedonian zone, immediately west of the Rhodope, in volcanic and some sedimentary rocks (Kondopoulou & Westphal 1986; Pavlides *et al.* 1988; Westphal *et al.* 1991) and in other parts of western Greece (Kissel & Laj 1988).

The directions obtained (Fig. 6) differ considerably from a geocentred dipole field ($D = 0°$; $I = 60°$) for many areas. When compared with the expected values for the region ($D = 8°$; $I = 57°$) using the Lower Oligocene pole for Eurasia (Westphal *et al.* 1986), a clockwise rotation of about 15° is noticed. A more important rotation of about 20° is found when the Oligocene pole for Africa is used ($D = 4°$; $I = 56°$).

Westphal (1993) suggested that the systematically low palaeolatitudes observed in Eocene to Oligocene formations from the Tethyan belt do not correspond to errors or large tectonic movements, but to an alternative pole position for Eurasia of 69°N and 215°E. This gives an expected direction at a site located at 41.5°N, 24.4°E of $D = 356°$ $I = 37.2°$. Such a direction would, however, imply an even larger rotation (about 26°).

The calculated mean inclination (around 50°) is shallower than expected compared with the present latitudinal position of the Rhodope, if the Lower Oligocene poles for Eurasia

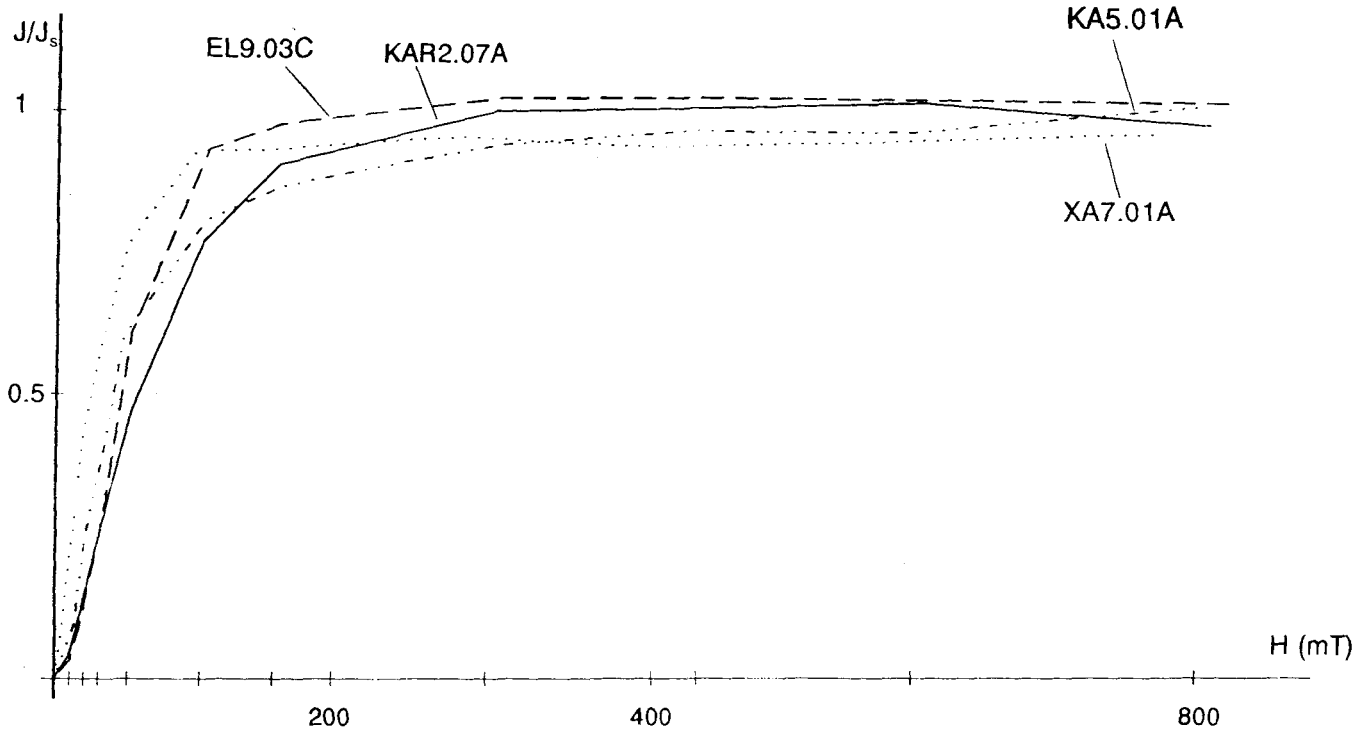


Figure 4. Examples of IRM curves from representative samples.

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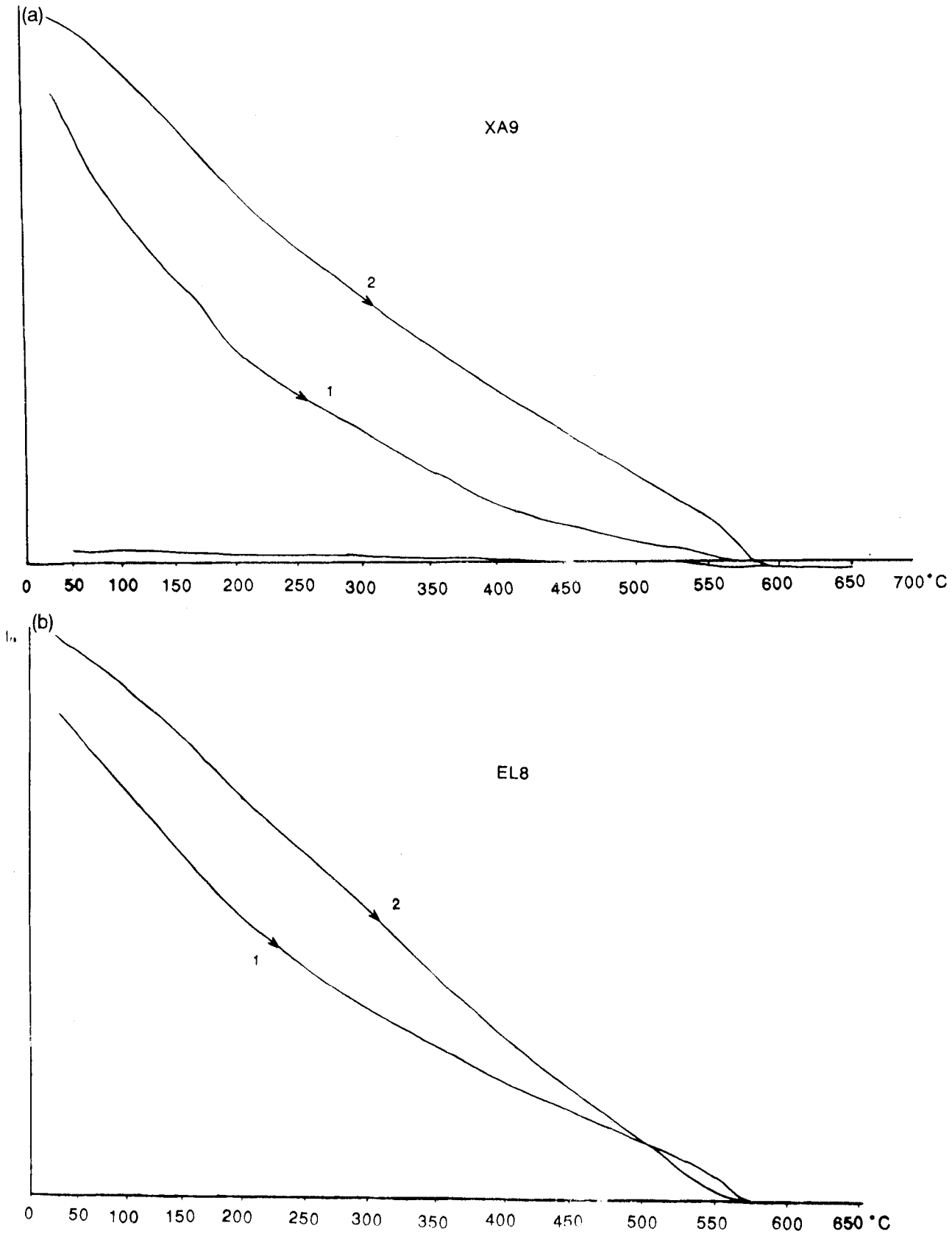


Figure 5. Examples of thermomagnetic analysis by study of the decrease of saturation demagnetization as a function of temperature for two samples: (a) granite; (b) volcanic. Vertical axis on arbitrary scale. 1, First heating curve; 2, Second heating curve.

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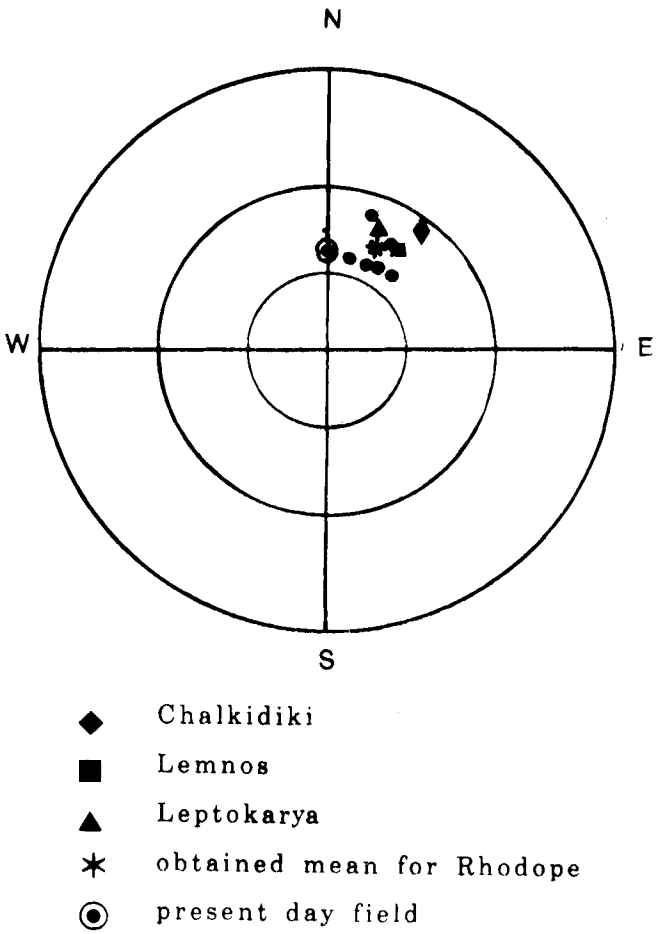


Figure 6. Stereographic projection of mean directions from studied areas (•) and from neighbouring regions.

and Africa (Westphal *et al.* 1986) are considered. Similar shallow inclinations had already been observed in earlier studies in Thrace (Kissel *et al.* 1986; Spais 1987) and in other areas of Greece (Kissel & Laj 1988; Pavlides *et al.* 1988). When, however, the alternative Oligocene pole for Eurasia of Westphal (1993) is taken into account, the present mean I value is steeper than expected.

In the first case (shallower inclinations) divergencies in inclination values may be the result of two factors.

(1) A more equatorial position of the area in Oligo-Miocene. Kissel & Laj (1988) suggested in fact a coherent evolution of the whole Aegean area during a latitudinal northward drift. Furthermore, they proposed that from the Early Paleocene to the Middle Miocene (60–15 Ma) the Aegean followed the latitudinal movements of Africa. This northward drift would imply lower inclinations than expected.

(2) A southward tilting of the formations studied, possibly as a result of imbrication due to north–south shortening in post Oligo-Miocene times. This contradicts all the geological evidence for this period (see also in the discussion).

In the second case (steeper inclinations) divergencies in inclination values may be the result of two factors.

(1) A secondary tilting of the magnetic vectors due to post-emplacement block movements (Beck 1992, and references therein). Blocks tilted to the north will have their vectors of magnetization also tilted downwards yielding apparently steeper I values. Tilting of blocks by rotation around subhorizontal axes is a common feature in extensional terrains.

(2) A less equatorial position of the area in the Oligo-Miocene. This case is considered as impossible (see the discussion).

A north–south post-Miocene extension affects the whole Aegean area, (McKenzie 1978), while a major NE–SW extensional regime was established in the Rhodope after the Late Eocene and was active through the whole of the Oligocene and Miocene; this extension was coeval with the magmatic activity in the Rhodope and perhaps genetically related to it (Maltezou & Brooks 1989; Kolokotroni 1992). A series of extensional domino faults or an extensional ramp system (Gibbs 1987) facing south may have been responsible for the observed high I values. Since tilting of the blocks was towards the north or north-east, the D values are not expected to be significantly affected by these movements. This likely mechanism of extensional faulting and block tilting could produce steeper I values than those predicted by the alternative pole position of Westphal (1993) for a site located at 41.5°N, 24.4°E. The relatively small scatter of the observed I values indicates, however, that tilting should have been quite uniform on the part of the Rhodope covered by this study which is not unlikely in a homogeneously extending terrain. If no northward drifting of the Rhodope is involved, then a *c.* 13° tilting of the blocks it comprises is implied.

The alternative Eurasian pole position of Westphal (1993) with a post-emplacement block tilting of about 13° seems to us a plausible explanation for the observed D and I values.

Cases which raised problems and certainly deserve additional work are described below.

(1) Vrondou granite. Two of the four sampling sites of this large plutonic body yielded results indicating no rotation. These sites are from the southern part of the main body and from the satellite outcrop of Eleon, both of which, according to Kolokotroni & Dixon (1991) and Kolokotroni (1992), have been subjected to intense flat-lying and wrench shearing contemporaneous with the intrusion. It is not known whether, and to what extent, this shearing could have affected the primary magnetization and whether a secondary magnetization was imposed on the already sheared granite after completion of its rotation. The other two sites on the Vrondou granite show a large scatter in both D and I values, but the directions of a persistent secondary component ($D = 29.8°$, $I = 52.3°$, $\alpha_{95} = 10.5°$) are consistent with directions obtained from the other apparently rotated granitic bodies in the area studied. A possible explanation is that during the period of intrusion stoping of magma had broken previously solidified parts of the pluton into blocks, which were then randomly rotated. The reheating of these blocks during the stoping was not sufficient to erase their primary magnetization, while their secondary magnetization imposed during the following cooling is the one that records the regional clockwise rotation.

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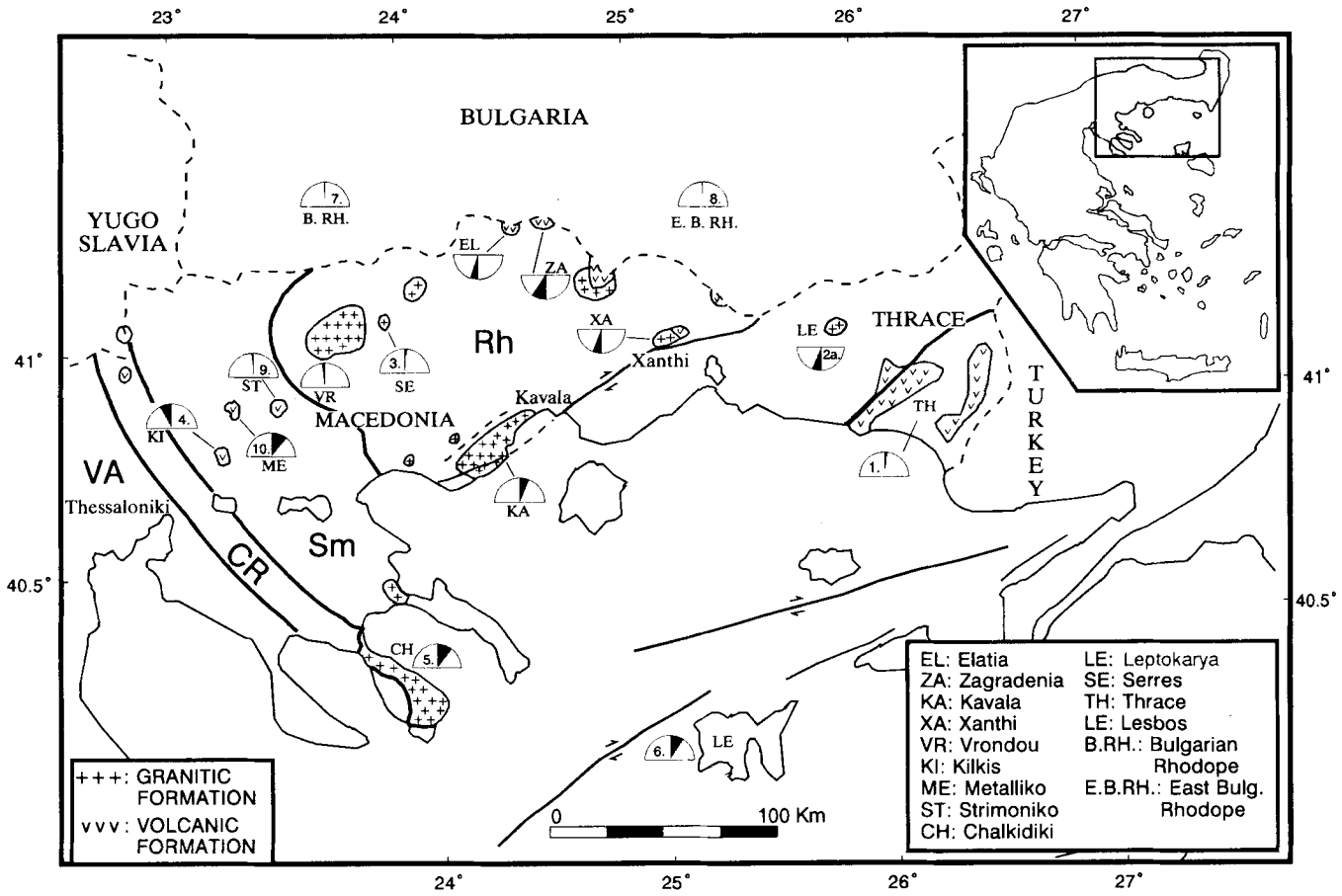


Figure 7. Published and new palaeomagnetic results of the north Aegean. (Numbers indicate the authors in Table 1.)

(2) Elatia granite. This granite is part of the large Scaloti intrusive complex, for which a K–Ar age of *c.* 38 Ma has been reported by Kyriakopoulos (1987). A whole-rock Rb–Sr age of 86 ± 27 Ma has been, however, reported for this same granite by Soldatos (1985). The crystallization age of the Elatia granite may indeed be Late Cretaceous, the Late Eocene K–Ar age indicating a re-equilibration by reheating above the closure temperature for this isotopic system and possibly a complete remagnetization of the granite.

DISCUSSION AND REGIONAL IMPLICATIONS

A discrepancy between the present results and those obtained from the Bulgarian Rhodope (Nozharov *et al.* 1990), where no Tertiary rotations have been found and the inclinations are as the present ones, requires discussion. We consider the possibilities that the inclinations vary due to tilting or to latitudinal movements.

In the first case, an unrotated and untilted Rhodopean domain has to be separated from a rotated and tilted one. If tilting and rotation are both linked to the same driving mechanism, then this mechanism could have affected only the southern (the Greek) part of the Rhodope, leaving the northern (the Bulgarian) part unrotated and untilted. This possibility is not further supported by tectonic or geomorphological evidence, but cannot be totally discarded at present.

In the second case (inclinations vary due to latitudinal movements), an important discontinuity has to separate two originally distant terrains (judging from the *I* values a separation of about 1000 km is implied). The present southern terrain (the Greek Rhodope) would have had to be drifted southwards over the present northern terrain (the Bulgarian Rhodope) in post Oligo-Miocene times. However, no geological evidence for such large-scale movements at this period exist. Important southwestward thrusts have been recognized in the Rhodope by Ivanov (1981, 1988), Papanikolaou & Panagopoulos (1981), Burgh *et al.* (1990) and Kilias & Mountrakis (1990); the age of the thrusting however, has been, estimated as Late Cretaceous to Eocene and thus cannot be related to the case considered here. Furthermore, in the Greek Rhodope the lower rather than the higher parts of the nappe stuck are exposed, while the striking similarities of the Tertiary evolution in the Greek and the Bulgarian Rhodope support their unity.

Alternatively, the discrepancy in the palaeomagnetic results for the Tertiary between the Greek and the Bulgarian Rhodope may not be real and may be the result of differences in the methodologies used. In fact, a careful evaluation of all published results on the Bulgarian Rhodope provides no evidence of a sufficient cleaning and component analysis. We suspect that future work may

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reveal post-Miocene clockwise rotations, at least in parts of the Bulgarian Rhodope.

A discrepancy between our results and the previously obtained ones in Thrace (Kissel *et al.* 1986; Spais 1987) supporting the idea of an unrotated Rhodope, needs further investigation. This area, belonging to the Eastern Rhodope, comprises the previously mentioned granite of Leptokarya in its westermost part, close to our sampling area. This granite displays a direction with $D = 22.8°$, $I = 46.4°$, $\alpha_{95} = 4.5°$ which implies a *c.* 15° to 27° clockwise rotation according to the VGP position adopted. If this is also the Westphal (1993) position, then a similar reasoning concerning the I values could be applied since I_{obt} is steeper than I_{exp}. With the present state of knowledge, we suggest that the wider area, which was affected by clockwise rotations after the Oligo-Miocene comprises the Serbomacedonian massif and the western Rhodope, with perhaps the eastern Rhodope unrotated (Fig. 7).

A NW–SE right-lateral wrench shearing was already active in the area during the intrusion of the Vrondou granite in the Oligocene (Kolokotroni & Dixon 1991; Kolokotroni 1992) and this might have contributed to a small extent to the rotations of the earlier Tertiary intrusions. The main rotation, however, was the result of a strain field which could be still active in the area.

Active tectonics in the north Aegean are characterized by right-lateral strike-slip movements coupled with a mainly NE–SW extension. The strike-slip component results from the westward movement of the Anatolian block, bounded from the north by the North Anatolian strike-slip fault (NASSF). This movement started not earlier than the Middle Miocene. The north Aegean is a zone of distributed dextral shear (McKenzie & Jackson 1983; Pavlides *et al.* 1990; Taymaz, Jackson & McKenzie 1991; Jackson, Haines & Holt 1992) and thus an area where rotations are likely.

Taymaz *et al.* (1991) proposed a model according to which the zone of dextral strike-slip plus north–south extension in the north and central Aegean is due to the splay out towards the south of the NASSF west of about 31°E. This splay out to several parallel faults is necessitated by the resistance of mainland Greece, supported by the continental collision between north-western Greece–Albania and the Apulia–Adriatic platform, to the purely strike-slip-accommodated westward movement of the Anatolian block. According to this model the southward splay out of the north Anatolian strike-slip system originally opened extensional basins in the Aegean, and was then accommodated by the overriding of the southern Aegean margin over the Mediterranean oceanic crust in the Hellenic Trench. Roll back of the subducting slab made the north–south extension self-sustained. How far north this extensional plus strike-slip regime extends and whether it includes part or the whole of the Rhodope is, however, not well constrained. Taymaz *et al.* (1991) suggest that it may extend as far north as 43°N, including therefore Thrace and the whole Rhodope, which in that case cannot be considered as the stable northern edge of the Aegean extensional domain. They also suggest that the same rotational—or no rotational—pattern exists for Thrace and the north Aegean islands. This hypothesis is further supported by the present results (clockwise rotation of about 20° in the Greek Rhodope) which are comparable with palaeomagnetic results in the north Aegean island of Lemnos, where a 25° clockwise rotation was measured in Neogene volcanics (Westphal & Kondopoulou 1989).

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